

ATARNotes

MATH METHODS 1&2

UNIT 1 REVISION

Presented by:
BECKY

OVERVIEW

1st content block:

- Linear review and coordinate geometry
- Set and function notation
- Functions
- Quadratics

2nd content block:

- Solving inequalities
- Hyperbola and truncus
- Square root functions, circles

3rd content block

- domain and range
- inverse functions

WELCOME

- Hi! I'm Becky
- I graduated in 2015
- Took a gap year
- Currently studying Architectural Design @ Monash (plus some Maths subjects because why not?)
- Love dogs. Here's mine...

WELCOME



He's such a good boy

POLL 1

- Why did you choose Methods?
- Have you been enjoying Methods so far?
- Which calculator do you use?

LINEAR REVIEW

A linear relation is a polynomial of degree one

$$y = \underline{mx} + \underline{c}$$

- c is the y-intercept
- m is the gradient

Solve the following for x :

$$\rightarrow 2x + 3 = 2$$

$$2x = -1$$

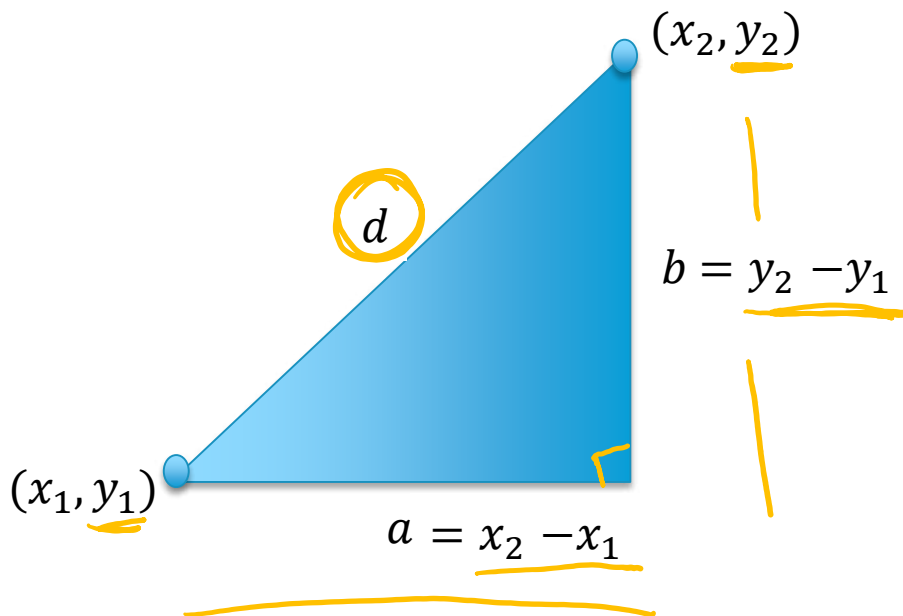
$$x = -\frac{1}{2}$$

Exam tip: It's good practice to box your answers!

DISTANCE

- To find the distance between two points, use the following formula

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$



$$a^2 + b^2 = c^2$$

$$(x_2 - x_1)^2 + (y_2 - y_1)^2 = d^2$$

$$d = \pm \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$d > 0$$

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

DISTANCE

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

- Key point: The distance formula is not in the formula sheet, so it must be memorised!

E.g. Find the distance between (0, 1) and (2, 4)

x_1 y_1 x_2 y_2

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$d = \sqrt{(0 - x_1)^2 + (1 - y_1)^2}$$

$$d = \sqrt{(0 - 2)^2 + (1 - 4)^2}$$

$$d = \sqrt{(-2)^2 + (-3)^2}$$

$$d = \sqrt{4 + 9}$$

$$d = \sqrt{13}$$

MIDPOINT

- The midpoint of a line segment is the 'middle' point.
- We find this by 'averaging' two coordinates

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

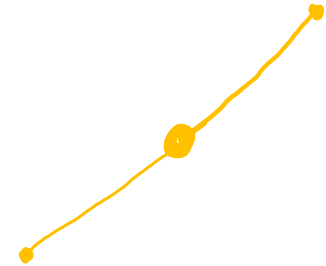
E.g. Find the midpoint between (1, 2) and (3, 4)

$$M = \left(\frac{1 + x_2}{2}, \frac{2 + y_2}{2} \right)$$

$$M = \left(\frac{1 + 3}{2}, \frac{2 + 4}{2} \right)$$

$$M = \left(\frac{4}{2}, \frac{6}{2} \right)$$

$$\mathbf{M = (2, 3)}$$



THE GRADIENT

The gradient of a line is referred to as $\frac{\text{rise}}{\text{run}}$.

$$m = \frac{\text{change in } y}{\text{change in } x} = \frac{y_2 - y_1}{x_2 - x_1}$$

Find the gradient of the line that passes between the points (1, 2) and (3, 4).

x_1 y_1 x_2 y_2

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$m = \frac{4 - 2}{3 - 1}$$

$$m = \frac{2}{2}$$

$$m = 1$$

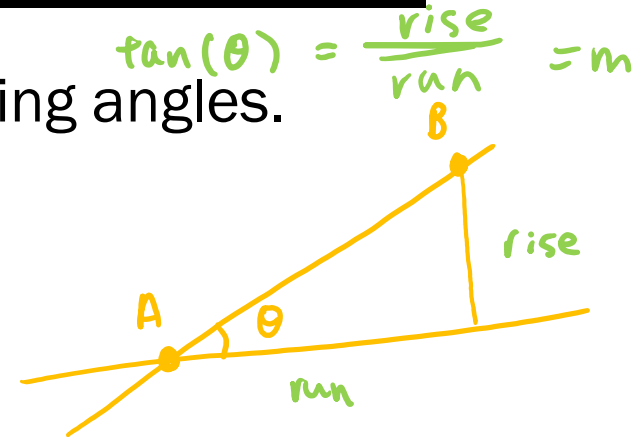
Note: You can use this to find the gradient between any two points on a graph (and hence the equation for the line)

THE GRADIENT

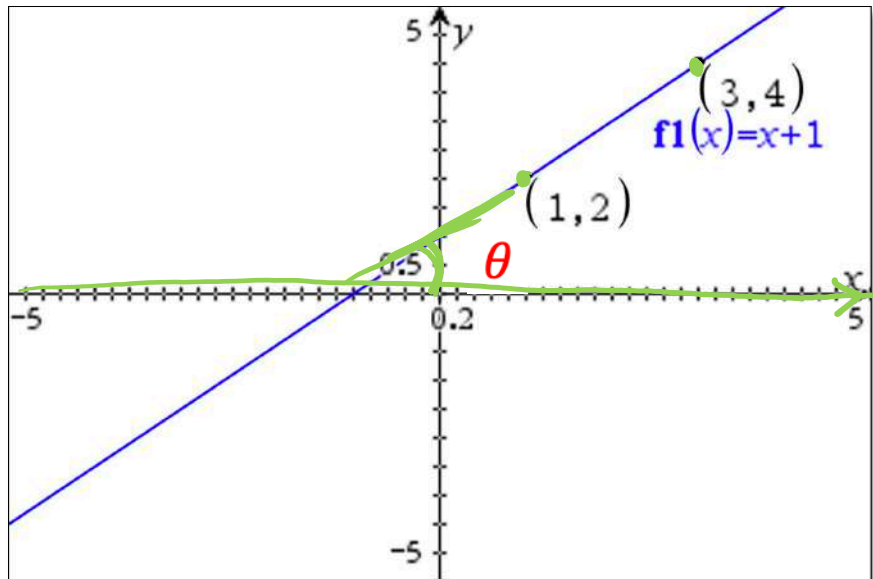
The gradient can also be represented using angles.

$$\tan(\theta) = \frac{\text{rise}}{\text{run}}$$

$$\theta = \tan^{-1}\left(\frac{\text{rise}}{\text{run}}\right)$$



Where θ is the angle from the positive direction of the x -axis.



$$\theta = \tan^{-1}\left(\frac{4 - 2}{3 - 1}\right)$$

$$\theta = \tan^{-1}(1)$$

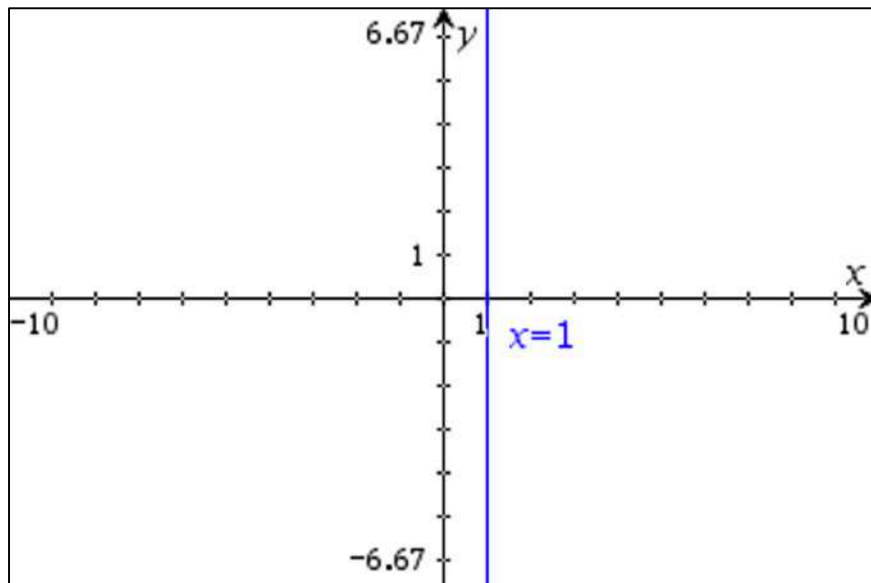
$$\theta = \frac{\pi}{4}$$

$$\theta = 45^\circ$$

VERTICAL AND HORIZONTAL LINES

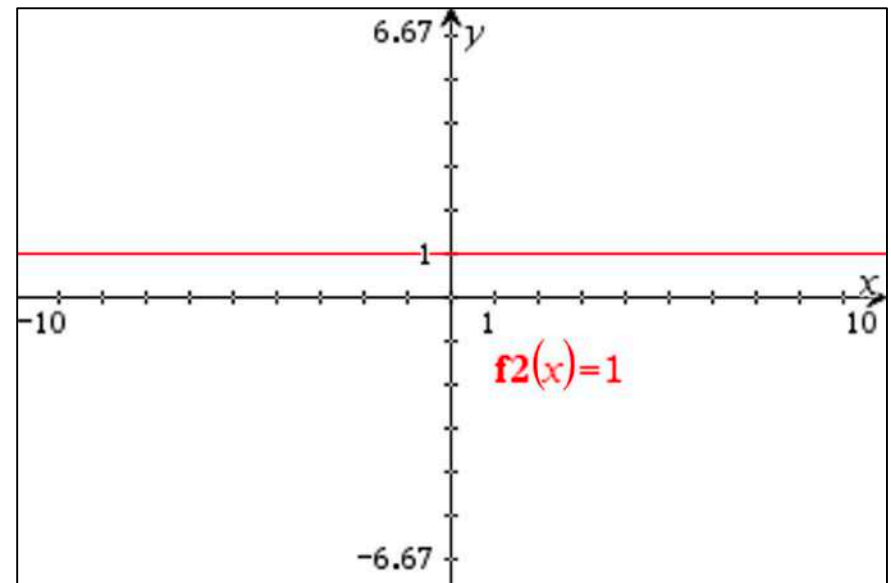
Vertical lines

- Expressed as $x = a$
- For any value of y , x will always equal 'a'



Horizontal lines

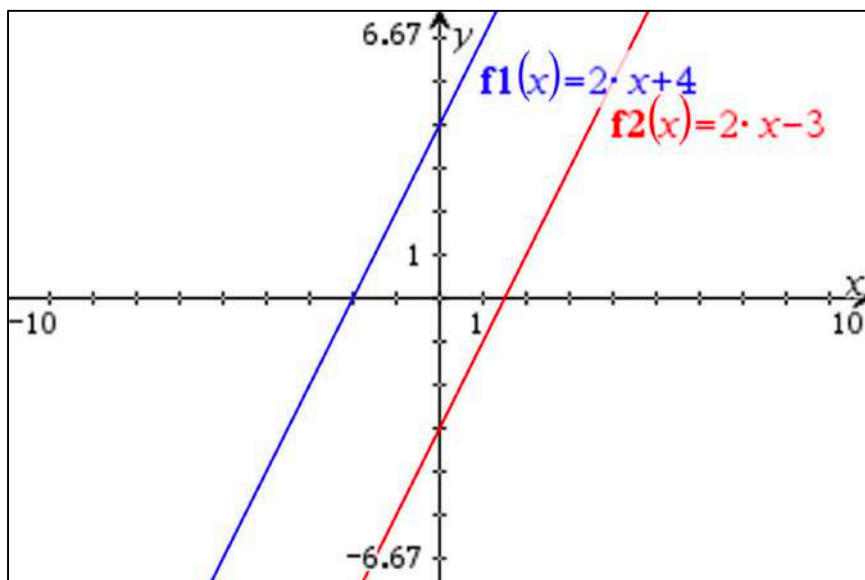
- Expressed as $y = b$
- For any value of x , y will always equal 'b'



PARALLEL AND PERPENDICULAR

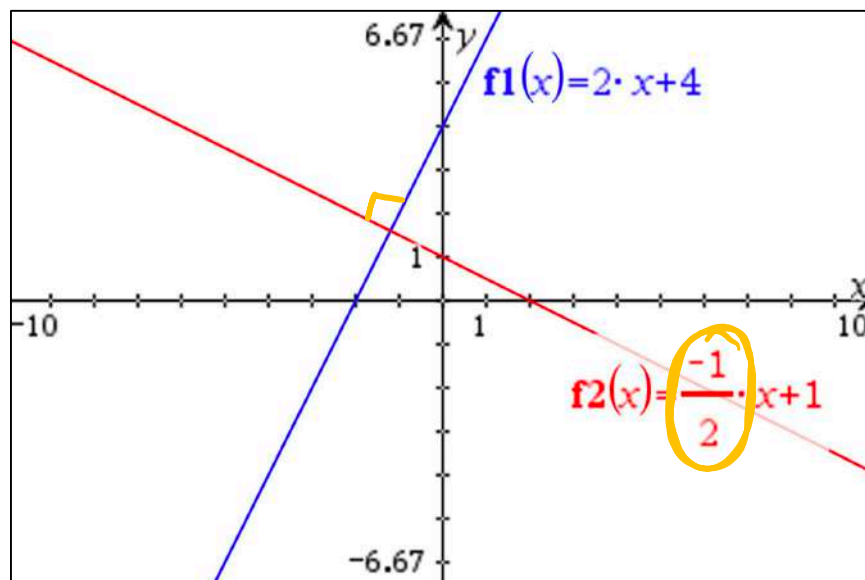
Parallel lines

- Same gradient (no intersect)
- $m_1 = m_2$
- E.g. $y = 2x + 4$ and $y = 2x - 3$



Perpendicular lines

- Cross at right angles
- $m_p = -\frac{1}{m}$
- E.g. $y = 2x + 4$ and $y = -\frac{1}{2}x + 1$



LINEAR EQUATIONS

- To find the equation of linear lines, you need two pieces of information

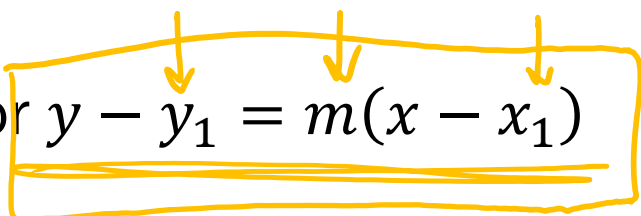
It's either

- Two points that lie on the line

Or

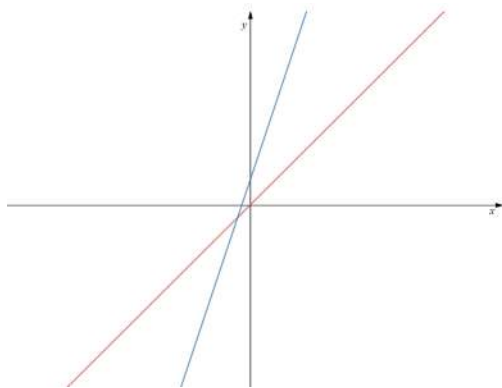
- Gradient of line and a point that lies on the line

Use either $y = mx + c$ or $y - y_1 = m(x - x_1)$

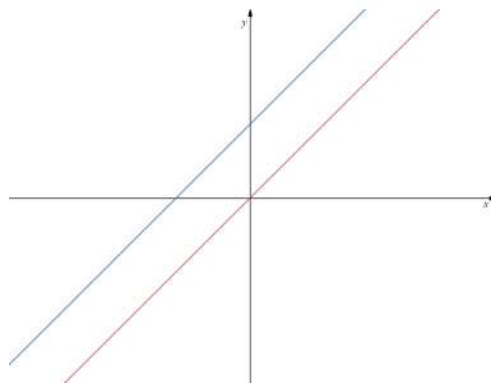
A hand-drawn yellow box encloses the point-slope formula $y - y_1 = m(x - x_1)$. Three yellow arrows point downwards from the top of the box to the variables y , y_1 , and x respectively. The box is drawn with a slightly irregular, hand-drawn style.

LINEAR EQUATIONS

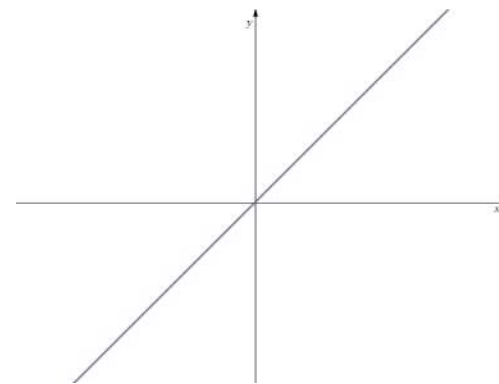
- Linear equations can have one, none or infinitely many intersections.



1 intersection



0 intersection



∞ intersections

- If gradients are the different, there will always be 1 intersection.

LINEAR EQUATIONS

- If the gradients are the same and y -intercepts are different, then they are parallel and will never intersect.
- If the gradients are the same and y -intercepts are the same then there will be ∞ many intersections.
- It is important that you remember/understand this because questions relating to these concepts come up in literally every single methods VCAA exam.

VCAA 2010 EX 2

The simultaneous linear equations ^① $(m-1)x + 5y = 7$ and ^② $3x + (m-3)y = 0.7m$ have infinitely many solutions for

A. $m \in \mathbb{R} \setminus \{0, -2\}$

B. $m \in \mathbb{R} \setminus \{0\}$

C. $m \in \mathbb{R} \setminus \{6\}$

D. $m = 6$

E. $m = -2$

$$\textcircled{1} \quad y = \frac{1}{5} \left(-(m-1)x + 7 \right) = \boxed{\frac{1-m}{5}}x + \boxed{\frac{7}{5}}$$

$$\textcircled{2} \quad y = \frac{1}{m-3} \left(-3x + 0.7m \right) = \boxed{\frac{-3}{m-3}}x + \boxed{\frac{0.7m}{m-3}}$$

$$\frac{1-m}{5} = \frac{-3}{m-3}$$

$$\frac{7}{5} = \frac{0.7m}{m-3}$$

$$m = -2 \quad \text{or} \quad \boxed{6}$$

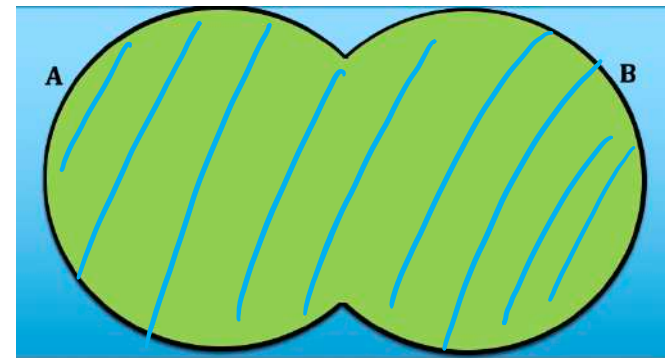
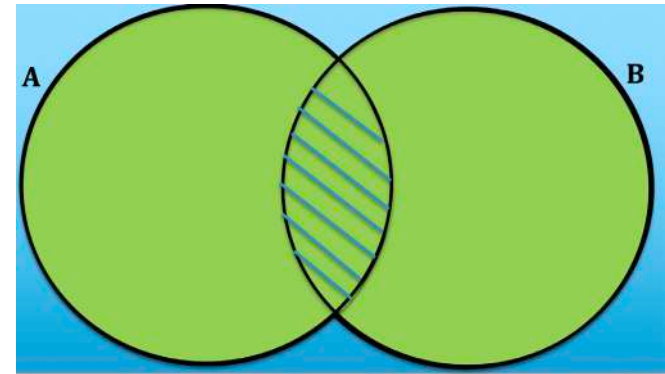
$$\boxed{m = 6}$$

NOTATION

- In the next few slides we will be looking at notation.
- Notation is really important because ultimately it is the language of mathematics so it is really important that you remember and know it.
- We will be looking at set notation and function notation. In other words, it is the way we represent a group of numbers and functions in mathematics.

SET NOTATION

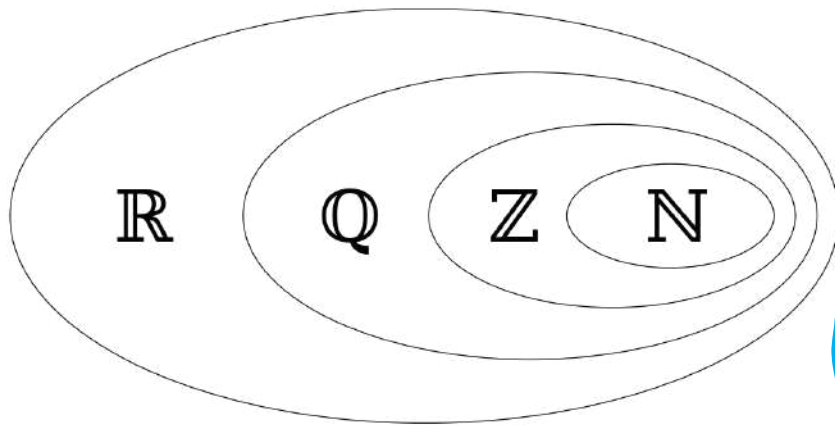
- A **set** is a collection of objects
- The intersection
 - Set of elements common to two sets
 - $\cap$: 'and'
- The union
 - Set of elements that are in both sets
 - $\cup$: 'or'
- Excluding
 - Denotes a set excluding an element
 - Symbol '\'
- Element of
 - Objects in set members of the set
 - Symbol ' $\in$ '



Key tip: Set notation is essential for representing domains and ranges in questions.

SET NOTATION

Some pre-determined sets are,



- **R**: Real numbers (include all rational numbers)
- **Q**: Rational numbers (number that can be in the form p/q where p and q are integers and q is not zero)
- **Z**: Integers (whole numbers, including negative, positive and zero)
- **N**: Natural numbers (positive whole numbers, sometimes zero)

SET NOTATION

- How do we represent a set of numbers?
- We use $\{ \}$ to denote either a particular value or particular values.
- E.g. the set of even numbers from 1 to 10 is $\{2,4,6,8,10\}$

INTERVAL NOTATION

Round or square brackets?

- Round brackets are used to indicate up to but excluding the end point.

e.g. $x > 4$ corresponds to $x \in (4, \infty)$ 

- Square brackets are used to indicate up to and including the end point.

e.g. $1 < x \leq 50$ corresponds to $x \in (1, 50]$ 

FUNCTION NOTATION

A function is defined by its rule and domain:

$$f: \underline{[a, b]} \rightarrow R, f(x) = y$$

$\mathbb{R} \setminus \{0\}$

$[0.4) \cup (7.6]$

- The first f is to state that we are defining as a function
 - $[a, b]$ is the domain (range can be calculated using the domain)
 - The second R is the codomain which we will **always observe to be R**
 - $f(x)=\dots$ is the rule of the function
- The best way to think of a function is as a machine. It has an input (the x -value) and an output (the y -value).
 - E.g. $f: R \rightarrow R, f(x) = 2x + 1$

Exam Tips

- Be careful when labelling the function name and dependent variable
- Look at marks to see if domain is required in your answer

FUNCTIONS

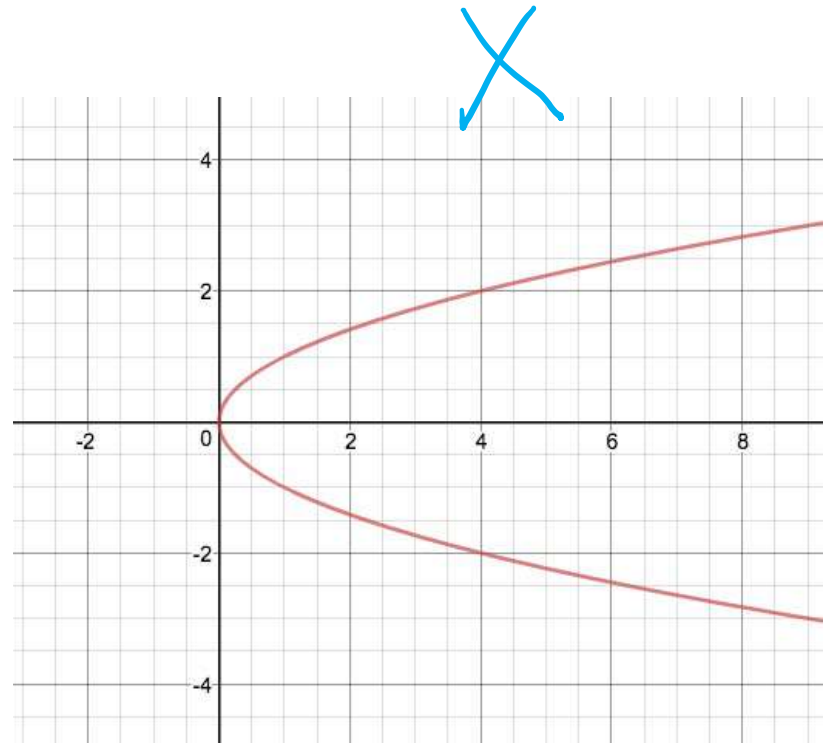
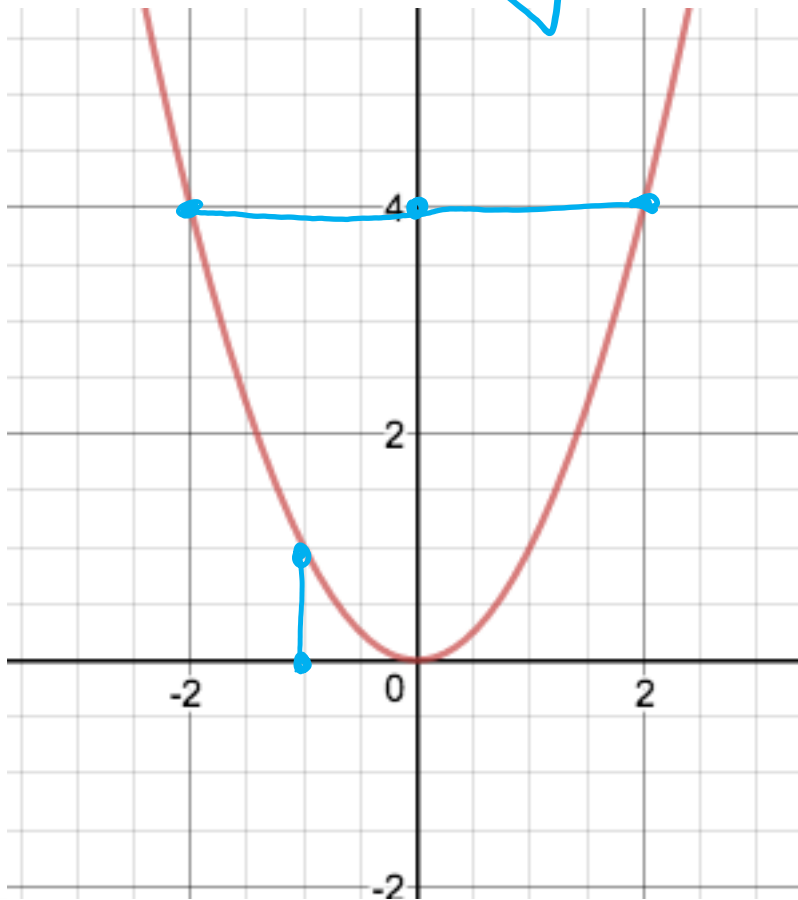
For a function to exist, its rule **MUST** pass the vertical line test. Sometimes we need to restrict the domain to be able to have a function.

Every x value has a unique y value

We use a vertical line test to check this.

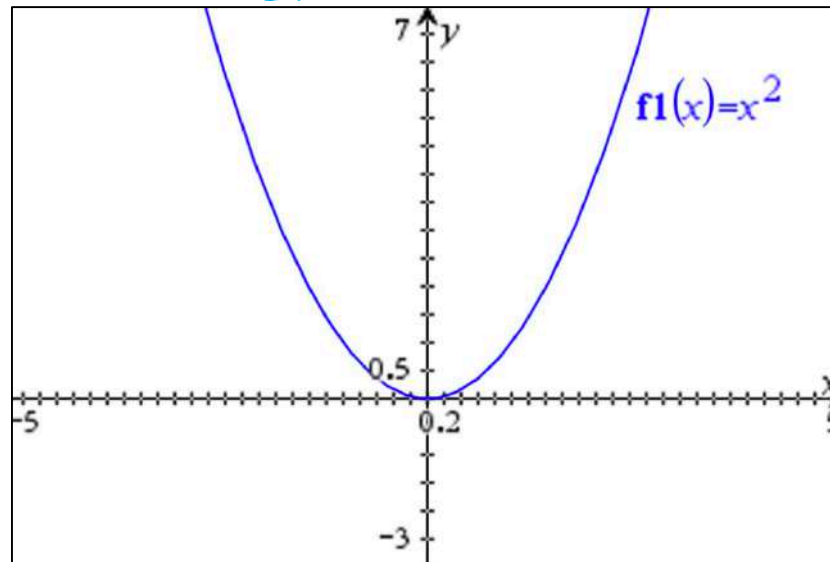
FUNCTIONS

Which of these graphs are functions?



QUADRATICS

- A quadratic is an equation of the second degree (x^2)
- The general form is: $f: R \rightarrow R, f(x) = ax^2 + bx + c$



- Turning point form: $f(x) = a(x - h)^2 + k$, where TP is (h,k)
- ➔ Factorised form: $f(x) = a(x - m)(x - n)$, where x-intercepts are at (m,0) and (n,0)

QUADRATICS

Quadratics (continued)

- To find the x-intercepts of a function we can use the **Null Factor Law**:

$$(a) \times (b) = 0$$

- Therefore, either a or b, or both must equal 0
- To apply the NFL, we need to first factorise the function using factorisation rules

QUADRATICS

The Quadratic Formula

- The Quadratic Formula allows us to find the x-intercepts of a quadratic *without* factorising

$$a = 2 \quad b = -3 \quad c = 4$$

For a quadratic of the form: $ax^2 + bx + c$,

$$x_{\text{intercept/s}} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Exam tips

$$\frac{-(-3) \pm \sqrt{(-3)^2 - 4(2)(4)}}{2(2)}$$

- Factorising is generally faster, use this for hard quadratics
- This formula is NOT in the formula sheet, so you must memorise it!
- Use brackets when subbing in a, b and c

QUADRATICS

- Quadratic equations come in a few different forms

Turning point form

$$y = a(x - h)^2 + k$$

$\Rightarrow (h, k)$ is the turning point

x -intercept form

$$y = a(x - m)(x - n)$$

$\Rightarrow (m, 0)$ & $(n, 0)$ are x - ints

The general form

$$y = ax^2 + bx + c$$

DISCRIMINANT

The Discriminant

$$\pm \sqrt{b^2 - 4ac}$$

- The discriminant allows us to find the number of solutions of a quadratic

$$\Delta = b^2 - 4ac$$

- There are three options:
 - $\Delta > 0$: Two solutions
 - $\Delta = 0$: One solution
 - $\Delta < 0$: No solution
- **Key point:** Before solving a quadratic, it's always good to test the number of solutions with the discriminant
- NB: The discriminant is derived from the quadratic formula!

$$x_{\text{intercept/s}} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

POLL 2

- Where does the distance formula come from?
- Is $\mathbb{Z} \subset \mathbb{N}$?

ATARNotes

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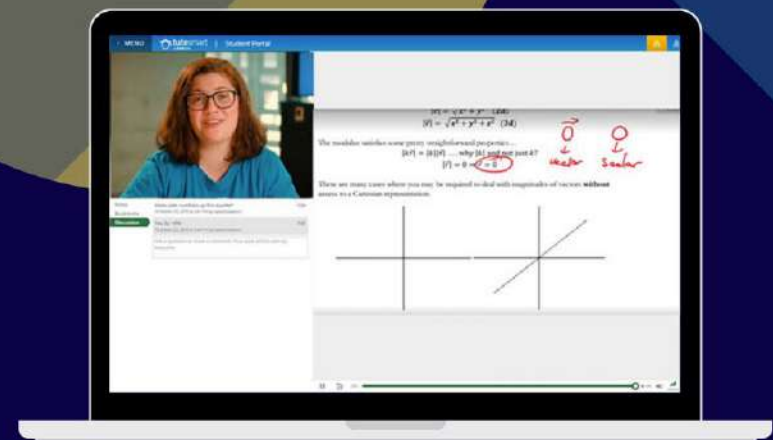
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LINEAR INEQUATIONS

- While a linear *equation* notes when two things are equal, an *inequation* shows one of four relationships:
 - ✂ — $a \geq b$ (Greater than or equal to)
 - — $a > b$ (Greater than)
 - — $a < b$ (Less than)
 - ↪ — $a \leq b$ (Less than or equal to)
- Remember: the “open”/”larger” side faces the greater quantity

LINEAR INEQUATIONS

$$\rightarrow (x)^2 < \frac{1}{2}$$

$$(-x)^2 < \frac{1}{2}$$

Rules for linear inequations

$$\rightarrow x < \frac{1}{\sqrt{2}}$$

$$-x < \frac{1}{\sqrt{2}}$$

$$x > -\frac{1}{\sqrt{2}}$$

- Generally the same rules as equations, except for:

$$-\frac{1}{\sqrt{2}} < x < \frac{1}{\sqrt{2}}$$

- If you multiply or divide by a negative number, you must **flip** the inequality (ex. $a > b$ becomes $-a < -b$)
- Some strange behaviour with complicated functions
- Ex: $x^2 < \frac{1}{2}$ becomes $-\frac{1}{\sqrt{2}} < x < \frac{1}{\sqrt{2}}$ when solving for x
- For these, it's best to draw a graph

NON-LINEAR INEQUALITIES

- Whenever you deal with linear inequations you can solve it straight forwardly with algebra
- However, when you deal with non-linear inequations, you cannot solve it straight forwardly with algebra, you need to solve it graphically.

NON-LINEAR INEQUATIONS

Solve for x :

$$x^2 \leq 16$$



$$(x)^2 \leq 16$$

$$(-x)^2 \leq 16$$

$$x^2 - 16 \leq 0$$

$$x \leq 4$$

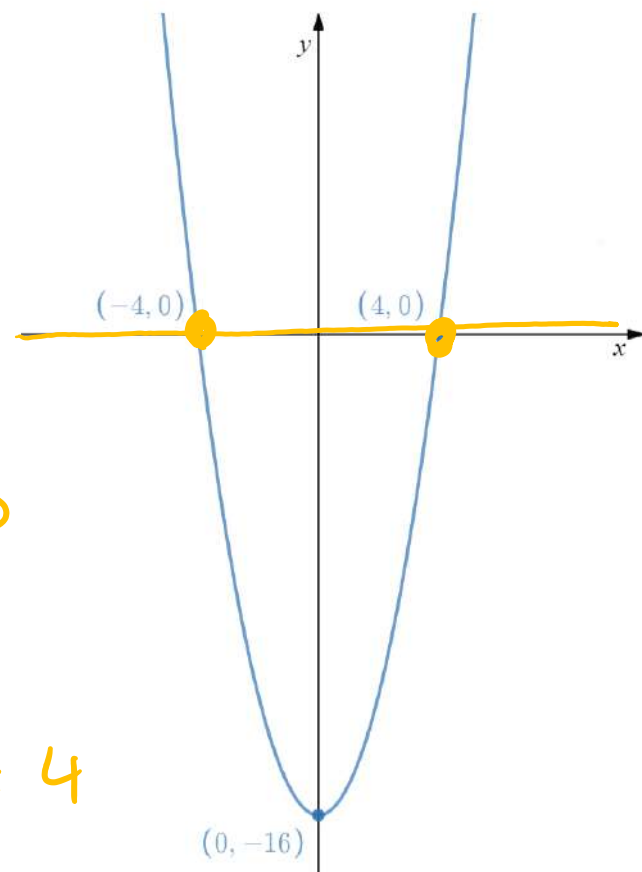
$$x \geq -4$$

$$-4 \leq x \leq 4$$



$$x^2 - 6 \leq 10$$

$$x^2 \leq 16$$



VCAA 2007 EX 2

$$\Delta = b^2 - 4ac$$

$$y = ax^2 + bx + c$$

$\Delta > 0$: two ~~solutions~~

The graph of $y = kx - 3$ intersects the graph of $y = x^2 + 8x$ at two distinct points for

A. $k = 11$

B. $k > 8 + 2\sqrt{3}$ or $k < 8 - 2\sqrt{3}$

C. $5 \leq k \leq 6$

D. $8 - 2\sqrt{3} \leq k \leq 8 + 2\sqrt{3}$

E. $k = 5$

$$x^2 + 8x = kx - 3$$

$$x^2 + 8x - kx + 3 = 0$$

$$\underline{x^2} + \underline{(8 - k)x} + \underline{3} = 0$$

$$(8 - k)^2 - 4(1)(3) > 0$$

$$k < 8 - 2\sqrt{3} \quad \text{or} \quad k > 8 + 2\sqrt{3}$$

RATIONAL FUNCTIONS

Rational functions are just any function which can be written as a fraction with both the *numerator* and *denominator* as polynomials

Remember: A **polynomial** is just a sum of x 's raised to integer powers.

Example of a polynomial: $2x^3 + 2x - \frac{5}{2}$

Examples of rational functions: $f(x) = \frac{7x^2}{3x+1}$ or $f(x) = \frac{2}{x^2-1}$

RATIONAL FUNCTIONS

Because you have a denominator which has x 's in it, sometimes you can run into problems with the domain!

Remember: **you cannot divide by zero**

Consider, what would happen if your function is $f(x) = \frac{1}{x-2}$

When $x = 2$, we'll end up with $f(2) = \frac{1}{2-2} = \frac{1}{0}$

How is this possible?

Answer: It's not!

RATIONAL FUNCTIONS

In order to avoid these divisions by zero, we have to make sure the *domain* excludes any values of x which cause issues

$$\text{If } f(x) = \frac{p(x)}{q(x)}, \text{ then } q(x) \neq 0$$

Example: If $f(x) = \frac{1}{x+3}$, $\text{dom } f = \underline{R \setminus \{-3\}}$

RATIONAL FUNCTIONS

Luckily in Methods, we only have to deal with two families of rational functions:

Hyperbolae, which are transformations of $f(x) = \frac{1}{x}$

Example: $f(x) = \frac{2}{x+1} + 2$

Trunci, which are transformations of $f(x) = \frac{1}{x^2}$

Example: $f(x) = \frac{3}{(x-1)^2}$

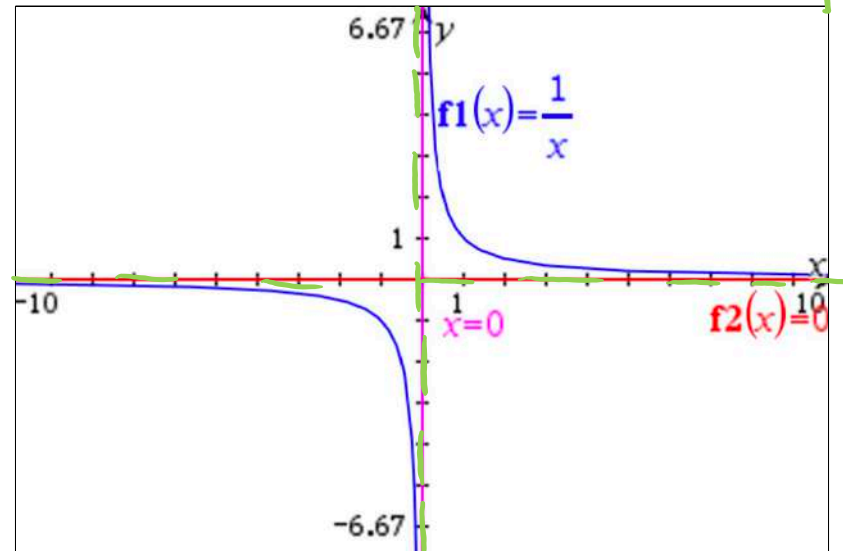
TYPES OF FUNCTIONS

Hyperbolas

- The most common form of a hyperbola is

$$f(x) = \frac{1}{x}$$

$$\frac{1}{f(x)} = 0$$



- The lines $x=0$ and $y=0$ are known as **asymptotes**
 - This means the function always approaches but **never touches** the vertical line $x=0$ and the horizontal line $y=0$

HYPERBOLAS

Hyperbolas (continued)

- Why can't $f(x) = \frac{1}{x}$ ever touch $x=0$ and $y=0$?

Rule 1: You cannot divide by zero!

$$f(0) = \frac{1}{0}$$

$$\begin{aligned}\text{Thus for } f_1(x) &= \frac{a}{x-h} \\ x-h &\neq 0 \\ x &\neq h\end{aligned}$$

$\therefore$ Asymptote at $x = h$

Rule 2: 1 cannot equal 0!

$$f(x) = 0$$

$$\frac{1}{x} = 0$$

$$1 = x \times 0$$

$$1 = 0$$

Impossible!

$$\begin{aligned}\text{Thus for } f_2(x) &= \frac{a}{x} + k \\ f_2(x) &\neq k\end{aligned}$$

$\therefore$ Asymptote at $y = k$

HYPERBOLAS

Hyperbolas (continued)

- For a hyperbola of the general formula

$$f(x) = \frac{a}{x - h} + k$$

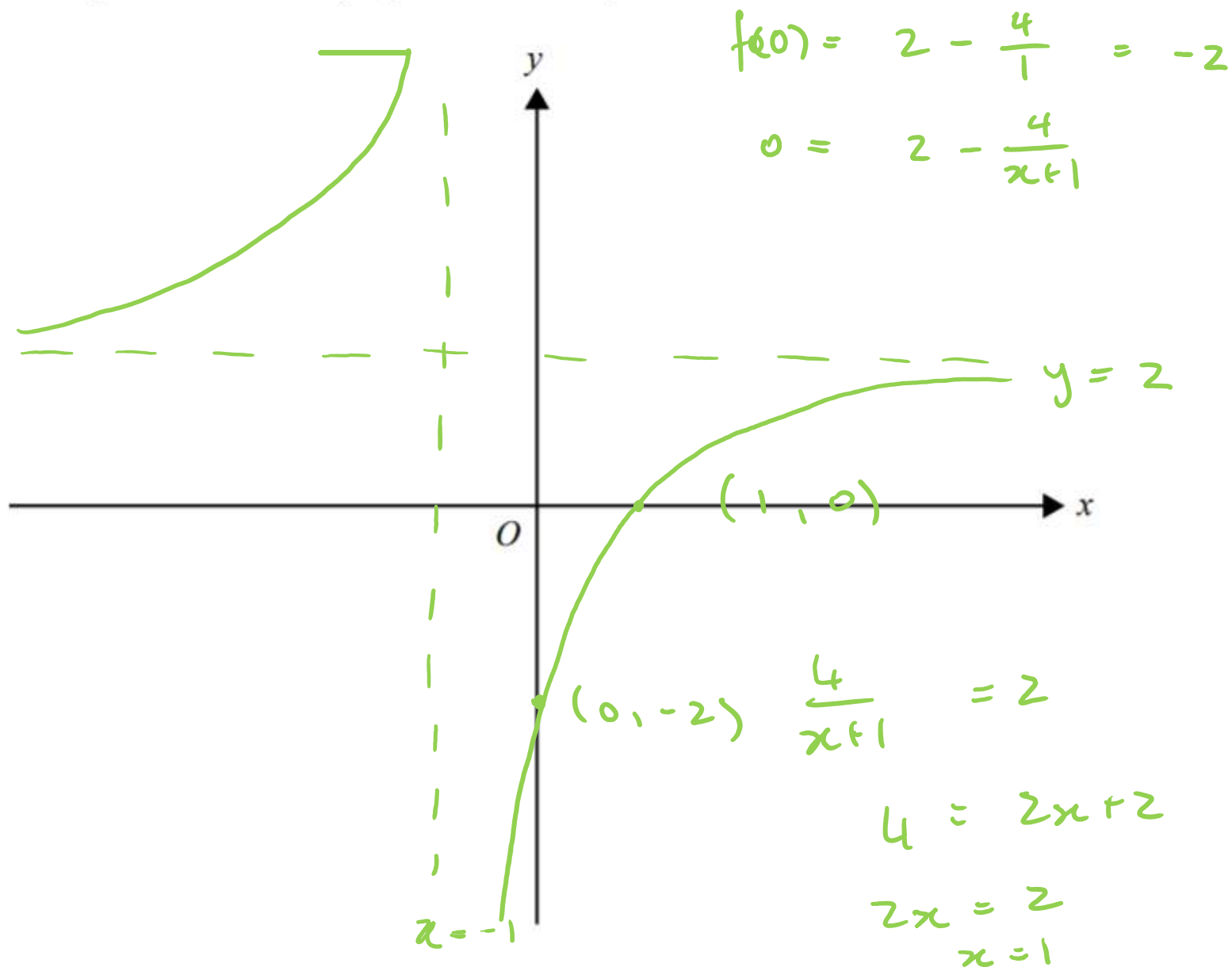
Asymptotes exist at:

- $x = h$
and
- $y = k$

Exam tips:

- Don't write asymptotes as $x \neq h$ or $y \neq k$ (always write them in the form $x = h$ and $y = k$)

On the axes below, sketch the graph of $f: \mathbb{R} \setminus \{-1\} \rightarrow \mathbb{R}$, $f(x) = 2 - \frac{4}{x+1}$.
 Label all axis intercepts. Label each asymptote with its equation.



HYPERBOLAE

Worked example

Sketch the graph of $y = \frac{x+2}{x+1}$, showing all intercepts and asymptotes

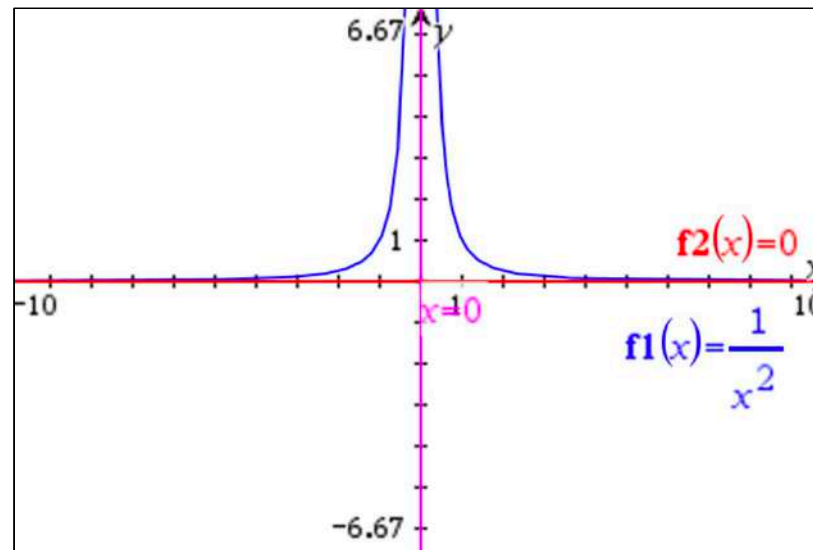
$$\begin{aligned} \frac{x+2}{x+1} &= \frac{x+1-1+2}{x+1} = \frac{\cancel{x+1}}{\cancel{x+1}} + \frac{-1+2}{x+1} \\ &= 1 + \frac{1}{x+1} \end{aligned}$$

TYPES OF FUNCTIONS

Truncus

- The most common form of a truncus is

$$f(x) = \frac{1}{x^2}$$



- It is always positive due to the x^2
- It **also** has asymptotes at $x=0$ and $y=0$

TRUNCUS

Truncus (continued)

- For a truncus of the general formula

$$f(x) = \frac{a}{(x-h)^2} + k$$

Asymptotes exist at:

- $x = h$

and

- $y = k$

Exam tips:

- Sketch asymptotes using a dotted line (and sketch the asymptote EVEN IF the asymptote is on the x or y axis)
- When sketching, your graph should NEVER touch the asymptote, and it should ALWAYS approach

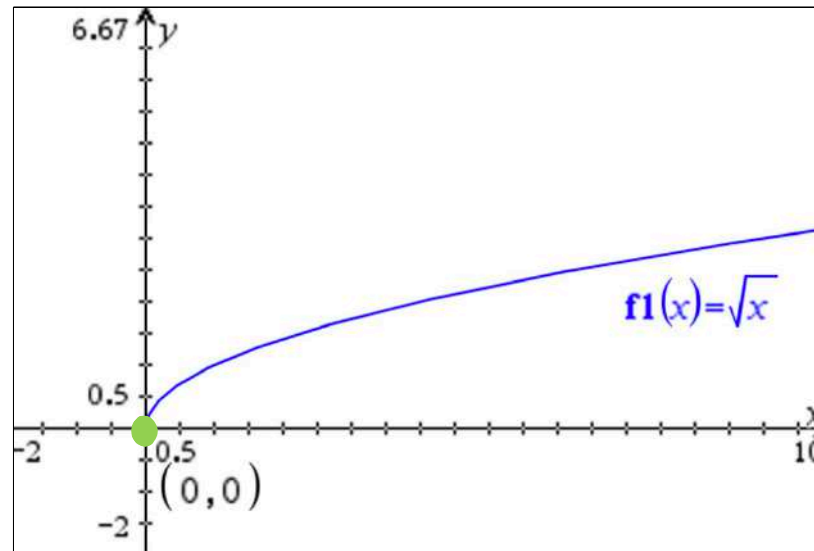
TYPES OF FUNCTIONS

Square root function

- The most common form of a square root function

$$f(x) = \sqrt{x}$$

$$\text{dom } f : [0, \infty)$$



- It starts at (0,0)
- It only travels in one direction, and always increases

SQUARE ROOT FUNCTIONS

Square root functions (continued)

- Why is $f(x) = \sqrt{x}$ unidirectional, and start at (0,0)?

Rule 3: You cannot square root negative numbers!

$$f(x) = \sqrt{x}$$

- You can square root zero (just not negative numbers)

$$x \geq 0$$

$\therefore f(x)$ will start at $x = 0$, and travel right

Thus for $f(x) = a\sqrt{x - b} + c$

$$x \geq b$$

And the start point will be (b, c)

NB: If, $f(x) = \sqrt{-x}$

$$-x \geq 0$$

$$x \leq 0$$

SQUARE ROOT FUNCTIONS

- Note that square root functions have end points
- They take on the general form

$$y = a\sqrt{b(x - \underline{h}) + \underline{k}}$$

Where

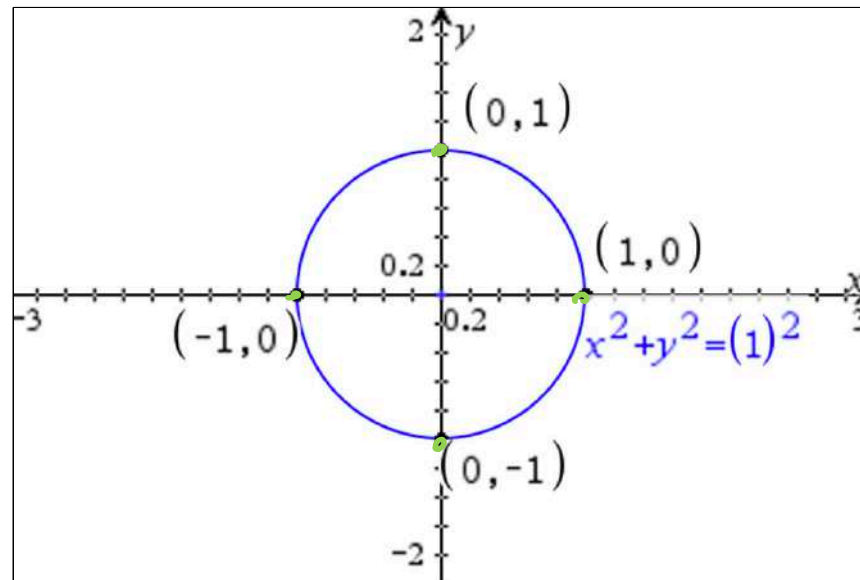
- (h, k) is the endpoint
- h and k are the horizontal and vertical translations
- If $a < 0$, the graph is reflected over the x axis
- If $b < 0$, the graph is reflected over the y axis

TYPES OF FUNCTIONS

Circles

- The general rule for a circle is

$$x^2 + y^2 = r^2$$



- The origin is (0,0)
- r is the radius and the circle is centered about the origin

CIRCLES

- The general equation of circles is

$$(x - h)^2 + (y - k)^2 = r^2$$

Where

- (h, k) is the centre
- r is the radius of the circle

POLL 3

- How many asymptotes does a hyperbola have?
- How many asymptotes does a truncus have?
- How many asymptotes does a square root function have?

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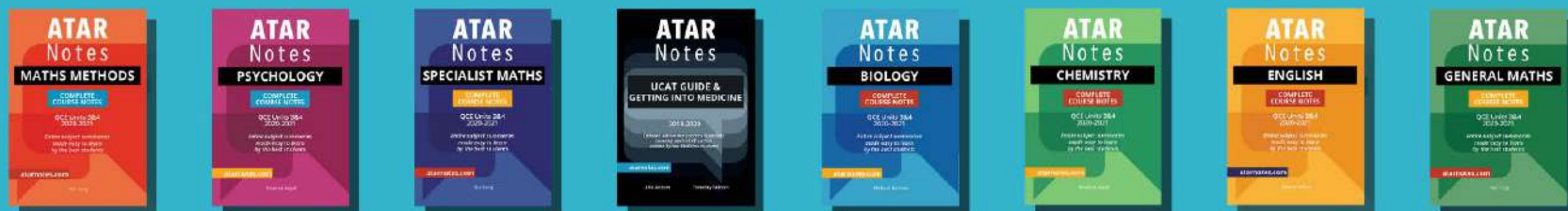


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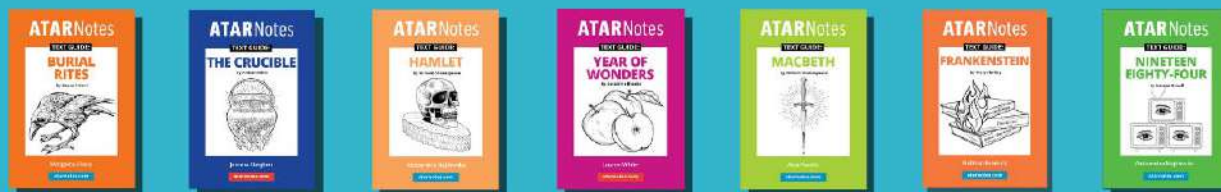
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ATARNotes

DOMAIN & RANGE

The domain of a function is a set of the x values which the function will work with. Basically the input to the function.

The range is the output of the function. All the y values that could possibly exist.

E.g. the domain of $f(x) = x^2$ is $\mathbb{R}$ and the range is $[0, \infty)$

Note, always a ~~curly~~ ^{round} bracket with infinity.



MAXIMAL DOMAINS

- Often we omit the $f: A \rightarrow B$ part and only write $f(x) =$ [rule]
- In this case, consider the maximal domain implied by rule

$$f(x) = \sqrt{x} \Rightarrow \text{dom } f = [0, \infty)$$

MAXIMAL DOMAIN

$$\sqrt{x}$$

The maximal domain of a function is the largest set of x-values in which the function exists.

EX: Find the maximal domain of $f(x) = \sqrt{x-1}$ and hence state its range.

$$x-1 \geq 0$$

$$x \geq 1$$

$$\text{dom } f : [1, \infty)$$

$$\text{ran } f : [0, \infty)$$

ROOT FUNCTIONS

Find the *maximal domain* of the following:

$$f(x) = \sqrt{3x - 2}$$

$$3x - 2 \geq 0$$

$$3x \geq 2$$

$$x \geq \frac{2}{3}$$

$$\text{dom } f : \left[\frac{2}{3}, \infty \right)$$

CHALLENGE QUESTIONS

Find the *maximal domain* of the function $f(x) = \sqrt{1 - x^2}$

$$1 - x^2 \geq 0$$

$$-1 \leq x \leq 1$$

$$1 \geq x^2$$

$$\text{dom } f: [-1, 1]$$



$$(x)^2 \leq 1$$

$$(-x)^2 \leq 1$$

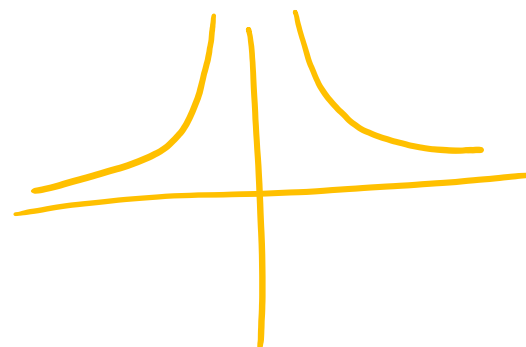
$$x \leq 1$$

$$x \geq -1$$

MAXIMAL DOMAIN

Find the maximal domain of the following relation and state the range

$$y = \frac{1}{x^2}$$



$$x^2 \neq 0$$

$$x \neq 0$$

$$\text{dom } f \rightarrow \mathbb{R} \setminus \{0\}$$

$$(-\infty, 0) \cup (0, \infty)$$

$$\text{ran } f: [0, \infty)$$

INVERSE FUNCTIONS

Consider $y = f(x)$

If we are given x , then we can easily find $y = f(x)$

- E.g. $y = f(x) = 2^x + 3$, and $x = 2$

What about the other way around?

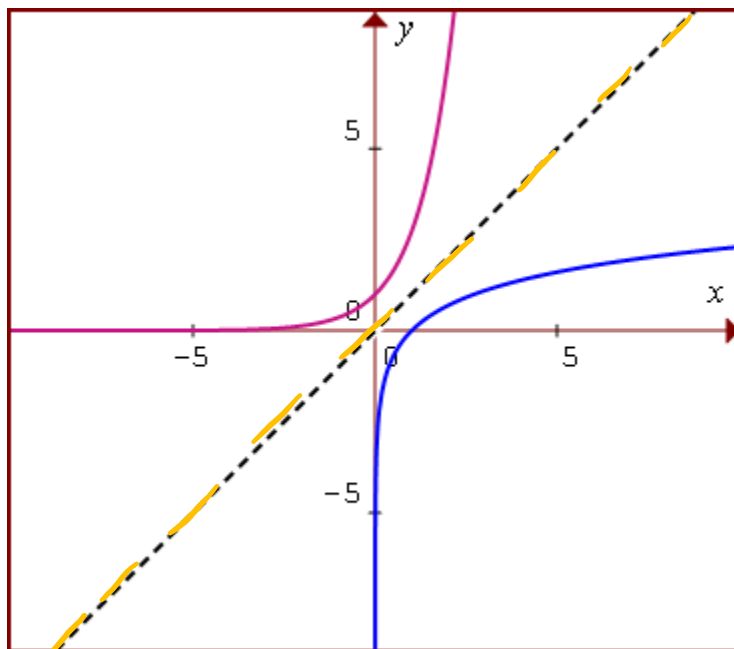
- If I give you y , can you find x ?

Inverse functions are when you want to *reverse* what a function has done!

INVERSE FUNCTIONS

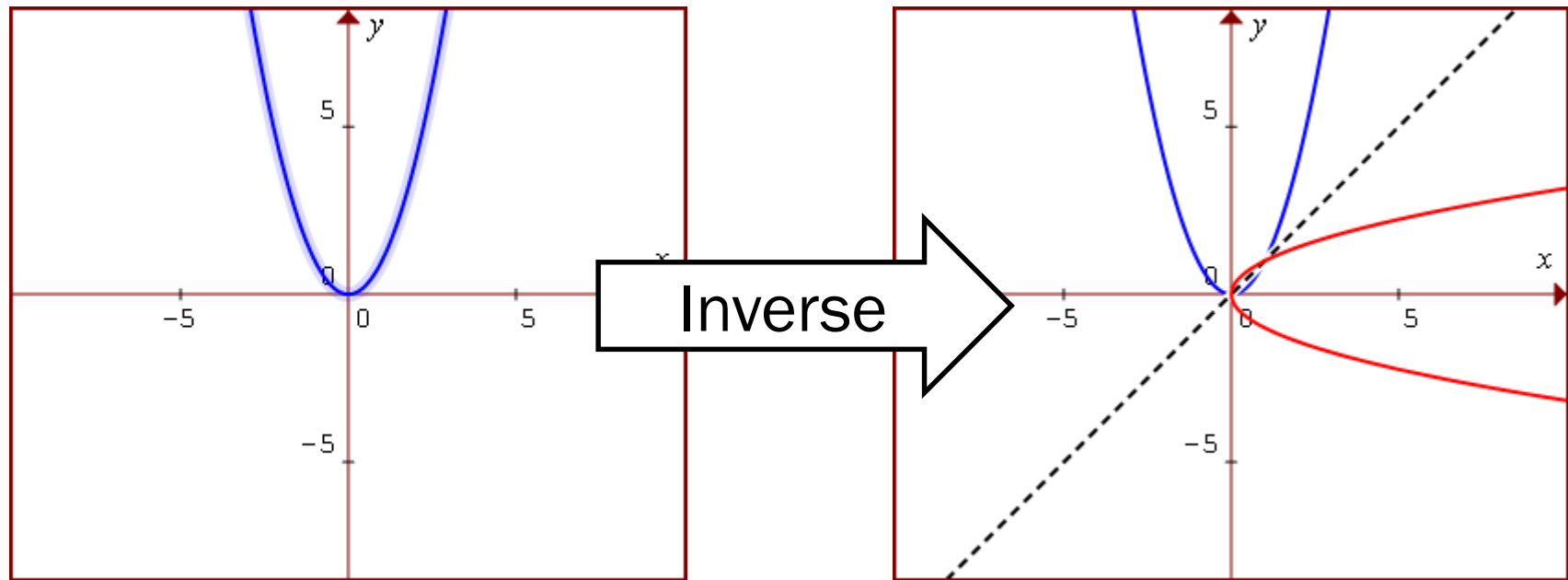
Geometrical meaning of the inverse

- The graphs of f and f^{-1} are related by reflection across $y = x$
- Note below both graphs are one-to-one



INVERSE FUNCTIONS

What if f is not one-to-one? We run into problems!



This is why the inverse only exists for one-to-one functions!

PROPERTIES THE INVERSE

$$(x, y) \rightarrow (y, x)$$

$$(x, x) \rightarrow (x, x)$$

- f and f^{-1} are related by a reflection about $y = x$
- $y = f(x)$ and $y = f^{-1}(x)$ intersect along $y = x$ (usually)
- Domain and range swap...
 - $\text{ran } f = \text{dom } f^{-1}$
 - $\text{dom } f = \text{ran } f^{-1}$

FINDING INVERSES

$$y = x^2$$

If f is a one-to-one function...

- Write $y = f(x)$
- Swap y and x , with explanation
- Rearrange for y as the subject

FINDING INVERSES

Example: $f(f^{-1}(x)) = x \Rightarrow x = 2f^{-1}(x) + 1$
Find the inverse of $f(x) = 2x + 1$, stating the domain and range.
 $f^{-1}(x) = \frac{1}{2}(x - 1)$

let $y = f(x)$, for inverse , swap x and y

$$x = 2y + 1$$

$$x - 1 = 2y$$
$$\underline{\underline{y = \frac{x - 1}{2}}}$$

$$\text{so } f^{-1}(x) = \frac{x - 1}{2}$$

FINDING INVERSES

Example: VCAA 2012

The rule for function h is $h(x) = 2x^3 + 1$. Find the rule for the inverse function h^{-1} .

let $y = h(x)$, for inverse , swap x and y

$$x = 2y^3 + 1$$

$$x - 1 = 2y^3$$

$$\frac{x-1}{2} = y^3$$

$$y^3 = \sqrt[3]{\frac{x-1}{2}}$$

$$\text{so } h^{-1}(x) = \sqrt[3]{\frac{x-1}{2}}$$

2 marks

FINDING INVERSES

Question 3

Marks	0	1	2	Average
%	9	34	56	1.5

$$h(x) = 2x^3 + 1, \text{ let } y = 2x^3 + 1$$

for inverse, swap x and y

$$\Rightarrow x = 2y^3 + 1, \text{ make } y \text{ the subject}$$

$$y = \sqrt[3]{\frac{x-1}{2}}, y \text{ is the inverse of } h(x)$$

$$h^{-1}(x) = \sqrt[3]{\frac{x-1}{2}}, x \in R$$



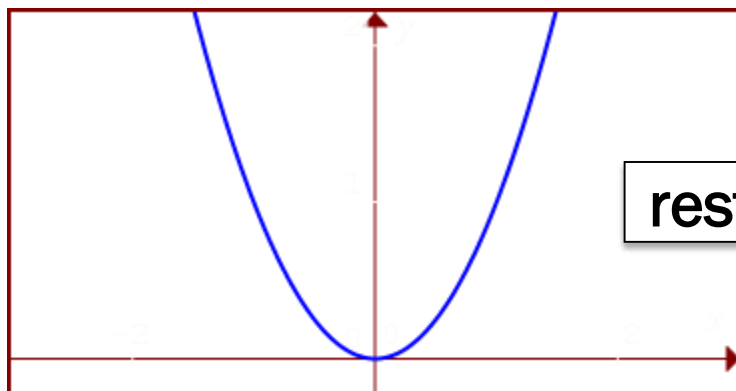
It is important that students do not proceed directly from $y = 2x^3 + 1$ to $x = 2y^3 + 1$. This is not correct working.

Students need to indicate that new working is starting. An approach is as shown above. A common error was to write

$$h^{-1}(x) = x = 2y^3 + 1.$$

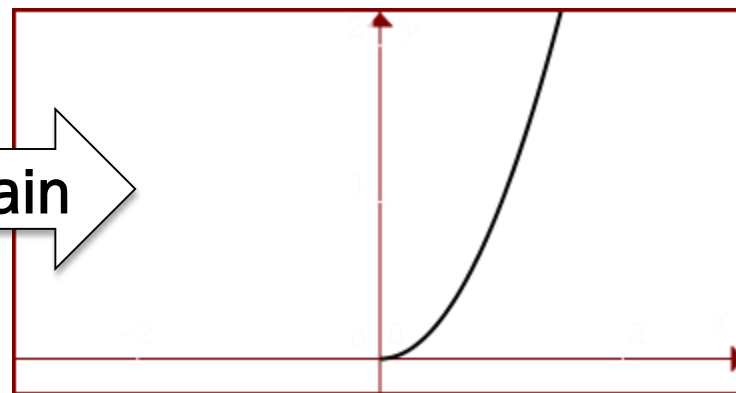
FINDING INVERSES

- If f isn't one-to-one, we can **restrict its domain** so it becomes a one-to-one function g .
- Since g is one-to-one, g^{-1} will exist



not one-to-one
 $\Rightarrow$ no inverse function

restrict domain



one-to-one
 $\Rightarrow$ inverse function
exists!

FINDING INVERSES

Example:

Let $f: [0, \infty) \rightarrow \mathbb{R}$, $f(x) = x^2 + 2$

Find the inverse of f , stating the domain and range. 3 mark

let $y = f(x)$, for inverse, swap x and y

$$x = y^2 + 2$$

$$y^2 = x - 2$$

$$y = \pm \sqrt{x - 2}$$

due to dom of f , we ~~take~~ take the positive arm

$$f^{-1}(x) = \sqrt{x - 2}$$

$$\text{dom } f^{-1} = \text{ran } f : [2, \infty) \quad \text{ran } f^{-1} = \text{dom } f : [0, \infty)$$

FINDING INVERSES

Example: VCAA 2014 Exam 2

Question 6

The function $f : D \rightarrow R$ with rule $f(x) = 2x^3 - 9x^2 - 168x$ will have an inverse function for

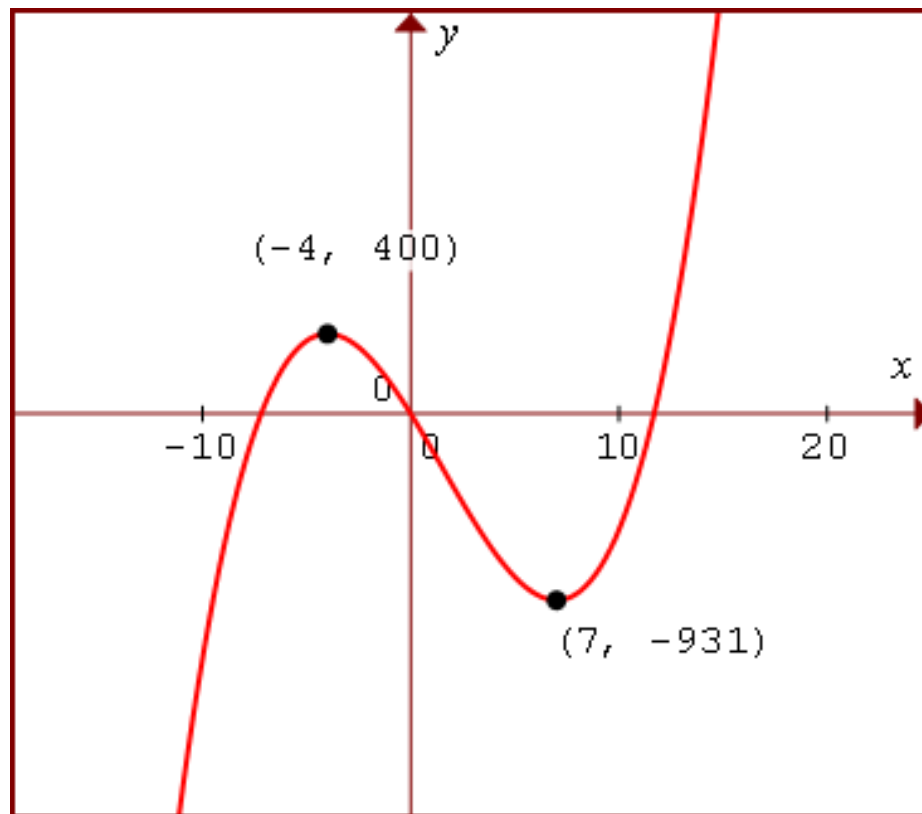
~~A.~~ $D = R$

B. $D = (7, \infty)$

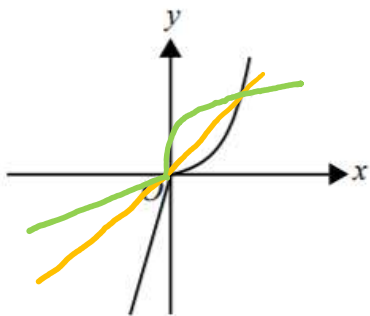
~~C.~~ $D = (-4, 8)$

~~D.~~ $D = (-\infty, 0)$

~~E.~~ $D = \left[-\frac{1}{2}, \infty\right)$

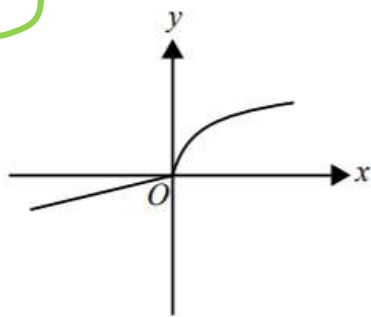


The graph of the function g with rule $y = g(x)$ is shown below.
(A one-to-one scale has been used.)

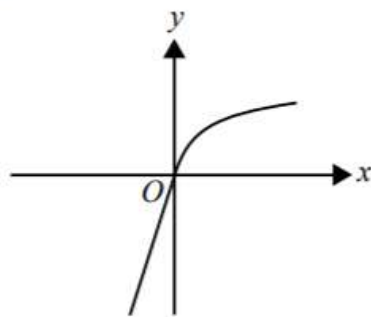


Which one of the following is most likely to be the graph of the inverse function of g ?
(A one-to-one scale has been used.)

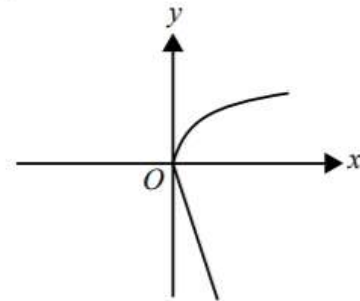
A.



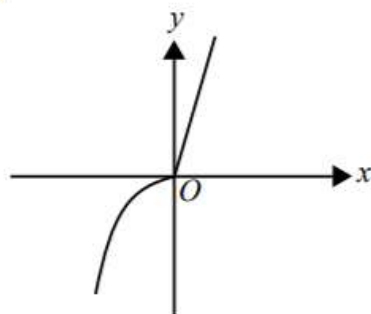
~~**B.**~~



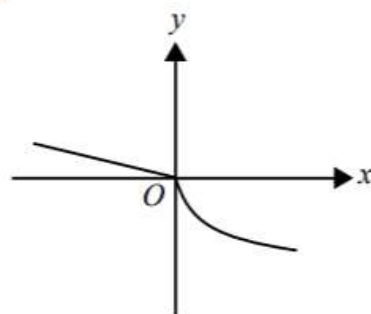
~~**E.**~~



~~**C.**~~



~~**D.**~~



VCAA 2009 EXAM 1

Question 3

Let $f: \mathbb{R} \setminus \{0\} \rightarrow \mathbb{R}$ where $f(x) = \frac{3}{x} - 4$. Find f^{-1} , the inverse function of f .

let $y = f(x)$, for inverse, swap x and y

$$x = \frac{3}{y} - 4$$

$$x - 4 = \frac{3}{y}$$

$$y = \frac{3}{x - 4}$$

$$\text{dom } f^{-1} = \text{ran } f : \mathbb{R} \setminus \{-4\}$$

$$\text{so } f^{-1}(x) = \frac{3}{x - 4}, \quad x \in \mathbb{R} \setminus \{-4\}$$

3 marks

VCAA 2011

The inverse function of $g: [2, \infty) \rightarrow R, g(x) = \sqrt{2x-4}$ is

~~A.~~ $g^{-1}: [2, \infty) \rightarrow R, g^{-1}(x) = \frac{x^2 + 4}{2}$

B. $g^{-1}: [0, \infty) \rightarrow R, g^{-1}(x) = (2x - 4)^2$

C. $g^{-1}: [0, \infty) \rightarrow R, g^{-1}(x) = \sqrt{\frac{x}{2} + 4}$

D. $g^{-1}: [0, \infty) \rightarrow R, g^{-1}(x) = \frac{x^2 + 4}{2}$

~~E.~~ $g^{-1}: R \rightarrow R, g^{-1}(x) = \frac{x^2 + 4}{2}$

— let $y = g(x)$, for inverse,
swap x and y

$$x = \sqrt{2y - 4}$$

$$x^2 = 2y - 4$$

$$x^2 + 4 = 2y$$

$$y = \frac{1}{2}(x^2 + 4)$$

INVERSE FUNCTIONS

- Remember that when we find inverse functions, we reflect them across the line $y = x$ to ascertain what it looks like visually

OR

- Algebraically, we swap the values of x and y .
- Think about what this means: we are swapping the values of x and y .
- That means the domain and range swap over.
- Also notice that if the point (a, b) lies on f , then on f^{-1} , this point is (b, a)
- Note that $f(f^{-1}(x)) = f^{-1}(f(x)) = x$

Let $f: R^+ \rightarrow R$ where $f(x) = \frac{1}{x^2}$.

a. Find $g(x) = f(f(x))$.

1 mark

b. Evaluate $g^{-1}(16)$, where g^{-1} is the inverse function of g .

1 mark

INVERSE FUNCTIONS

- We know that inverse functions are achieved by reflecting them across the line $y = x$.
- That means that at least one of the intersection points of f and its inverse, f^{-1} , occur on the line $y = x$.

- To find intersection points, you're probably tempted to set
$$f(x) = f^{-1}(x)$$

- However, algebraically this is a mess and often times you can't solve it.

- Instead you set

$$\underline{f(x) = x} \quad \text{or} \quad \underline{f^{-1}(x) = x}$$

(2,2)

VCAA 2011 EX 2

c. The cubic function q is defined by $q: R \rightarrow R, q(x) = 3 - 2x^3$.

i. Write down an expression for $q^{-1}(x)$.

let $y = q(x)$, for inverse, swap x and y

$$x = 3 - 2y^3$$

$$x - 3 = -2y^3$$

$$y^3 = \frac{3 - x}{2}$$

$$y = \sqrt[3]{\frac{3 - x}{2}}$$

$$\text{so } q^{-1}(x) = \sqrt[3]{\frac{3 - x}{2}}$$

VCAA 2011 (cont)

- ii. Determine the coordinates of the point(s) of intersection of the graphs of $y = q(x)$ and $y = q^{-1}(x)$.

$$3 - 2x^3 = \sqrt[3]{\frac{x-3}{2}}$$

$$3 - 2x^3 = x$$

$$2x^3 + x - 3 = 0$$

2 + 2 = 4 marks

POLL 4

- True or False: A function and its inverse will ALWAYS intersect on the $y = x$ line.

ATARNotes

QUESTIONS?